Lecture 27: Alternative Water Reactors: The CANDU Heavy Water Reactor

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 Introduction

While the Pressurized Water Reactor (PWR) has dominated the global commercial market using enriched uranium and a single-phase light water coolant loop, it is not the only viable path to commercial nuclear power. The CANadian Deuterium Uranium (CANDU) reactor represents the most successful alternative design philosophy in the world. By prioritizing neutron economy above all else, the CANDU design enables the use of unenriched (natural) uranium. This lecture explores the distinct physics, engineering trade-offs, economics, and geopolitical history of the heavy water, pressure-tube architecture.

2 Global Prevalence and History

Currently, there are **31 official CANDU reactors** operating worldwide, representing about 7% of the global nuclear fleet.

- **Canada (The Primary Fleet):** Canada operates the vast majority (19 units), heavily concentrated in Ontario, where they provide over 50% of the province's electricity grid.
- **The Export Market:** Canada has successfully exported genuine CANDU units to five countries: South Korea, Romania, China, Argentina, and Pakistan.
- **The Indian Fleet (IPHWRs):** India is actually the second-largest operator of this technology. Due to historical proliferation concerns (discussed later in this lecture), India was forced to proceed independently, reverse-engineering the Canadian design to build their own domestic fleet of approximately 18 Indian Pressurized Heavy Water Reactors (IPHWRs). If these unlicensed derivatives are included, the global footprint of heavy-water, pressure-tube reactors is closer to **50 units** ($\approx 11\%$ of the global fleet).

3 Physics: The Heavy Water Trade-off

CANDU uses Deuterium Oxide (D_2O) as the moderator.

- **Absorption Advantage:** $\sigma_a^H = 0.33$ barns vs. $\sigma_a^D = 0.0005$ barns. Deuterium captures almost no neutrons, preserving enough neutron population to maintain criticality with natural uranium.

- **Slowing Down Disadvantage (Quantitative):** Deuterium is twice as heavy as Hydrogen, making it less efficient at removing kinetic energy per collision.
  - Logarithmic Energy Decrement: $\xi_{H2O} \approx 0.92$ vs. $\xi_{D2O} \approx 0.51$.
  - Collisions to thermalize: Light Water ≈ 19; Heavy Water ≈ 35.
  - *Consequence:* Neutrons travel much further while slowing down. The lattice pitch must be large ($\approx 28 - 30$ cm) compared to a PWR fuel rod spacing of (≈ 1.2 cm).

4 Design: The Distributed Pressure Boundary

Because the heavy water lattice requires a massive volume, a single forged steel pressure vessel (like a PWR) would be prohibitively large and impossibly thick to manufacture. Therefore, CANDU reactors use a distributed pressure boundary.

4.1 The Calandria and Fuel Channel Geometry

To accommodate the large volume and separate the hot coolant from the cold moderator, the CANDU utilizes a "tube-within-a-tube" design across hundreds of horizontal channels.

- **The Calandria:** The heavy water moderator is kept in a large, horizontally oriented, low-pressure stainless steel tank called the Calandria. It is massive (roughly 6.0 meters long and 7.6 meters in diameter), but because it operates near atmospheric pressure, its walls only need to be about 1 inch thick.

- **The Fuel Channels:** The Calandria is pierced horizontally by 380 individual fuel channels. Each channel consists of:

  1. **Inner Pressure Tube (The Primary Boundary):** A thick Zirconium-2.5% Niobium pipe (≈ 104 mm inner diameter) holding the fuel bundles and high-pressure (10 MPa) primary coolant.
  2. **Outer Calandria Tube:** A thinner Zircaloy pipe (≈ 129 mm inner diameter) welded directly to the Calandria tank walls.
  3. **Annulus Gas:** The gap between the two tubes is filled with a continuously circulating insulating gas (like CO_2) to prevent heat loss from the hot pressure tube to the cold moderator. Garter springs keep the inner tube from sagging and touching the outer tube.

4.2 The Two-Fluid System: Where Does the Moderation Happen?

A common misconception is that the heavy water flowing directly over the fuel pins acts as the primary moderator. While some moderation occurs within the pressure tube, the volume is insufficient to thermalize neutrons efficiently. Unlike a PWR, the CANDU physically separates the coolant and the moderator into two isolated systems:

- **The Coolant (Inside the Tubes):** The D_2O flowing through the pressure tubes is hot ($\approx 310°$C), highly pressurized, and acts purely as a heat transfer fluid. Fast neutrons born in the fuel easily escape the tube with very little slowing down.

- **The Moderator (Outside the Tubes):** The actual moderation occurs in the Calandria tank, which is filled with a separate, massive volume of unpressurized, cold D_2O ($\approx 70°$C).

The Neutron Journey & Lattice Pitch: Because D_2O requires roughly 35 collisions to thermalize a neutron, the pressure tubes must be spaced far apart (a lattice pitch of $\approx$ 28.6 cm). A fast neutron is born in one bundle, escapes the pressure tube, bounces around in the cold D_2O of the Calandria until it thermalizes, and then drifts into a *neighboring* pressure tube to initiate a new fission. This allows the moderator to remain dense (maximizing neutron economy) while the coolant runs hot (maximizing thermodynamic efficiency).

4.3 The Fuel Bundles and Spatial Self-Shielding

Unlike the massive, 12-foot fuel assemblies of a PWR, CANDU fuel is packaged in short, compact bundles. A standard bundle is roughly 50 cm long, weighs 24 kg, and has an outer diameter of approximately 10.2 cm.

- **Internal Geometry:** 37 individual fuel elements are arranged in concentric rings (1 center pin, surrounded by rings of 6, 12, and 18).
- **Tight Tolerances:** The pins are thick ($\approx$ 13.1 mm). The 18 pins on the outer ring are packed so tightly that the physical gap between them is only 1 to 2 millimeters.
- **Bearing Pads (The "Runners"):** To allow these bundles to move through the reactor, the outer fuel pins are fitted with raised Zircaloy **bearing pads**. These act like the runners on a sled, maintaining a precise coolant gap against the pressure tube wall and allowing the bundle to slide smoothly without scratching the inner surface.
- **Spatial Self-Shielding (Flux Depression):** Because the outer ring forms a dense barrier of uranium, thermal neutrons diffusing in from the bulk moderator are heavily absorbed by the outer ring. The thermal flux at the center of the bundle is 20% to 30% lower than at the outer edge. Operators must keep the hottest outer pins below their thermal limits, leaving the inner pins operating well below potential capacity. Advanced designs (like CANFLEX) use thinner pins on the outside to flatten this power profile.

4.4 Continuous On-Line Refueling

Because natural uranium has very low excess reactivity, the reactor cannot operate on an 18-month batch cycle like a PWR. Fuel must be replaced constantly to maintain criticality.

- **The Mechanism:** Specialized robotic fueling machines latch onto opposite ends of a single pressure tube while the reactor is running at full power. One machine pushes fresh bundles in, which mechanically slides the entire string of bundles forward, pushing the spent bundles out the other side into a receiving machine.
- **Frequency:** This is a daily operation. In a standard CANDU 6, operators typically refuel **2 to 4 channels every single day**, inserting 4 to 8 fresh bundles per channel.

5 Economics: The Cost of Heavy Water

The single largest economic hurdle for the CANDU design is the staggering cost of the heavy water itself. Because deuterium is naturally present in only about 1 in 6,400 water molecules, extracting it requires massive, energy-intensive chemical separation plants.

- **Unit Cost:** Industrial bulk nuclear-grade heavy water typically costs between \$300 to \$800 per kilogram.
- **Capital Penalty (The "Fill-Up"):** A standard commercial CANDU requires roughly **500 metric tons** (500,000 kg) of heavy water to fill both the Calandria and the primary coolant loops. This means the heavy water alone adds approximately **\$150 million to \$400 million** to the upfront capital cost of building the plant.
- **The Capital vs. Fuel Trade-off:** The economics of a CANDU balance out over its 60-year lifespan. While a PWR gets its light water essentially for free, it locks the utility into paying millions annually for uranium enrichment. The CANDU pays a massive upfront capital penalty for the moderator, but benefits from incredibly cheap, unenriched natural uranium fuel for the life of the reactor.
- **Leak Recovery:** Because the fluid is essentially liquid gold, CANDU plants are meticulously engineered with vapor recovery systems (often utilizing cold silica gel beds) to catch, condense, and recycle nearly every drop of heavy water that evaporates from pump seals or valves.

6 Issues: Tritium & Efficiency

1. **Tritium Production & Economics:** Deuterium absorbs neutrons to become Tritium (^{3}H):

$$^2\text{H} + n \longrightarrow ^3\text{H} \quad (T_{1/2} = 12.3 \text{ years})$$

   - **The Hazard (Safety First):** Inside the reactor, neutron capture turns the D_2O into DTO (Deuterium Tritium Oxide). If this leaks into the plant atmosphere, rapid isotopic exchange with normal humidity converts it into HTO (Tritiated Water). Because it is chemically identical to water, HTO is a significant ingestion and skin-absorption hazard for workers. The primary reason utilities build massive detritiation facilities is to reduce this occupational radiation exposure, not purely to make a profit.
   - **The Production Rate:** A standard CANDU produces approximately **130 grams** of Tritium per year as the heavy water absorbs neutrons.
   - **The Income Stream:** Because global supply is incredibly tight, Tritium is highly valuable ($\approx \$30,000$/gram). A single reactor generates roughly **\$3.9 million** worth of Tritium annually. Centralized facilities (like the Darlington Tritium Recovery Facility in Ontario) harvest roughly 2 to 2.5 kg per year from the fleet, yielding up to **\$75 million** annually. This revenue helps offset the massive capital costs of the D_2O and the detritiation plant itself.
   - **The Market:** Once harvested, it is sold for:
     - Self-powered emergency lighting (exit signs, watch dials).
     - Medical and Life-science tracers.
     - Fusion Research (e.g., the ITER project).
     - Nuclear Weapons (Boost gas).

2. **Thermodynamic Efficiency:** The pressure-tube design and material limits constrain outlet temperatures to lower values than a light water PWR.

   - *Limitations:* The outlet temperature is restricted to $\approx 310°$C.

- *Result:* The thermodynamic efficiency is lower, typically in the range of 28% to 30% (compared to $\approx 33\%$ for a PWR).
- *Solutions:* One approach is to use other fluids in the coolant loop such as light water, but that requires at least some Uranium enrichment due to neutron absorption.

3. **Fuel Utilization vs. Burnup:** There is a critical distinction between *burnup* and *resource utilization.*
   - **Low Burnup:** Because CANDU fuel is natural uranium (0.7% ^{235}U), it can only sustain criticality for a short time. The discharge burnup is low ($\approx 7,500$ MWd/MTU) compared to a PWR using enriched fuel ($\approx 50,000$ MWd/MTU).
   - **High Waste Volume:** This means a CANDU produces $\approx 7\times$ the *volume* of spent fuel bundles compared to a PWR for the same energy generated.
   - **High Resource Efficiency:** However, because a PWR requires $\approx 8 - 10$ kg of mined uranium to produce 1 kg of enriched fuel (discarding the rest as tailings), the CANDU actually extracts *more energy per kg of mined uranium* than a PWR.

The Mass Balance Problem: Where does the ^{235}U go?

To understand why the CANDU has higher overall resource efficiency despite its low fuel burnup, we must look at the actual ^{235}U concentrations throughout the fuel cycle:

- **Natural Uranium Ore:** $\approx 0.71\%$
- **Enrichment Tailings (Depleted U):** $\approx 0.25\%$
- **Spent CANDU Fuel:** $\approx 0.2\%$
- **Spent PWR Fuel:** $\approx 0.8\%$ to 1.0%

The PWR Trap: To make 1 kg of 4% enriched PWR fuel, you must mine roughly 8 to 10 kg of natural uranium ore. The enrichment plant throws away the excess mass as "tails" containing roughly 0.25% ^{235}U. The reactor then burns the enriched fuel, but discharges it when it still has roughly 0.8% ^{235}U remaining (ironically, richer than the natural ore pulled out of the ground!). In a PWR cycle, you lose ^{235}U in two separate places.

The CANDU Advantage: The CANDU skips the enrichment process entirely (zero enrichment tails) and burns the natural uranium straight down to $\approx 0.2\%$ before discharge. Because it avoids enrichment losses and also benefits from breeding and burning ^{239}Pu along the way, the CANDU extracts about 15% to 20% more total thermal energy from the original ton of mined ore than a PWR.

The CANDU Waste Disadvantage: While the total mass of "leftovers" is similar for both cycles, the *radiological hazard* is vastly different. For every 10 kg of mined uranium, a PWR produces roughly 1 kg of highly radioactive High-Level Waste (spent fuel) and 9 kg of Depleted Uranium (DU) tailings. DU is only mildly radioactive, safe to handle, and has industrial/military uses (e.g., kinetic energy penetrators, radiation shielding). In contrast, the CANDU puts all 10 kg of that natural uranium through the core, meaning **100% of the initial mass becomes HLW** contaminated with fission products, requiring complex shielding, cooling, and ultimate deep geological disposal.

7 Geopolitics, Proliferation, and Market Economics: Why Aren't There More CANDUs?

Despite its elegant engineering and independence from uranium enrichment, the CANDU reactor ultimately lost the global market share battle to the Light Water Reactor (LWR). This was driven by a convergence of economics, standardization, and a massive geopolitical stigma.

1. **The Proliferation Vulnerability (On-Line Refueling):** The same feature that makes the CANDU operationally brilliant makes it a major proliferation risk. Because the reactor does not need to shut down to refuel, a rogue operator could theoretically slip a natural uranium bundle into the core, allow it to absorb neutrons to breed Plutonium-239, and push it out a few weeks later before it gets contaminated with unwanted Pu-240. In a PWR, breeding illicit plutonium requires shutting down the entire plant and opening the pressure vessel, which is immediately obvious to international inspectors.

2. **The Indian Reverse Engineering & The "Smiling Buddha":** The proliferation fears became reality in 1974 when India detonated its first nuclear weapon (Operation Smiling Buddha). The weapons-grade plutonium was bred inside CIRUS, a Canadian-supplied heavy-water research reactor based on the early CANDU technology.

    - **The Fallout:** In response, Canada immediately halted all nuclear cooperation with India.
    - **The IPHWR Fleet:** Cut off from Canadian support, Indian engineers successfully reverse-engineered the pressurized heavy-water architecture. They built a massive domestic fleet of unlicensed clones known as Indian Pressurized Heavy Water Reactors (IPHWRs).
    - **The Export Freeze:** The international community responded to the 1974 test by forming the Nuclear Suppliers Group (NSG) to heavily restrict nuclear exports. The CANDU design was cast under intense geopolitical suspicion by non-proliferation advocates, severely crippling its export market.

3. **The "VHS vs. Betamax" Standardization War:** While the CANDU was fighting proliferation stigmas, the Light Water Reactor was rapidly standardizing the global supply chain. Because Admiral Rickover chose the PWR for the US Navy submarine fleet, American industry scaled up the manufacturing, regulatory frameworks, and workforce around enriched water reactors. By the time the commercial market boomed, buying a PWR was simply easier, cheaper, and safer from an investment standpoint.

4. **Physical Scale and Complexity:** Because natural uranium has such a low probability of fission, the CANDU core must be physically massive to achieve the same thermal output as a tightly packed, highly enriched PWR core. A 1000 MWe CANDU requires a significantly larger containment building than a 1000 MWe PWR. Furthermore, the two incredibly complex, robotic fueling machines that routinely latch onto high-pressure radioactive pipes introduce significant mechanical complexity and maintenance costs compared to a PWR's static pressure vessel.

8 CANDU Safety and Containment Strategy

By eliminating the massive, ultra-thick steel pressure vessel used in PWRs and BWRs, the CANDU design fundamentally alters its safety profile and containment requirements.

1. **Distributed Pressure and Passive Heat Sinks:** The high-pressure coolant is distributed across roughly 380 to 480 individual channels rather than a single massive volume.
    - *Small Break LOCA:* A single pressure tube rupture only depressurizes one channel. Emergency core cooling systems can easily compensate, preventing a core-wide transient.
    - *The Calandria as a Heat Sink:* If a channel completely dries out, the overheating inner pressure tube will physically sag and touch the cooler outer calandria tube. The massive volume of 70°C heavy water in the calandria acts as a passive heat sink, drawing decay heat away from the fuel and significantly delaying or preventing a localized meltdown.
2. **The "Spaghetti" Problem and Header Breaks:** The trade-off for eliminating the large pressure vessel is the creation of thousands of individual mechanical connections.
    - *Feeder Pipes:* The face of the reactor is covered in a dense, complex web of feeder pipes that route coolant into and out of the individual channels.
    - *Large Break LOCA:* These tiny pipes connect to large distributor pipes called "headers." A rupture of a main header represents the CANDU equivalent of a Large Break LOCA, depressurizing a significant fraction of the core simultaneously.
    - *Positive Void Coefficient:* Unlike Light Water Reactors, a CANDU has a slightly positive void coefficient of reactivity. If a header breaks and coolant flashes to steam (voids), the formation of steam voids alters moderation and neutron leakage in a way that produces a net positive reactivity insertion. Consequently, CANDUs are engineered with **two fully independent, ultra-fast shutdown systems** (spring-assisted solid shutoff rods dropping from the top, and high-pressure liquid gadolinium poison injected from the side).
3. **The Vacuum Building Containment Solution:** Because a header break would still fill the reactor building with radioactive steam, CANDUs require robust containment. However, Canadian plants are typically built as massive multi-unit stations (e.g., the Bruce Nuclear Generating Station operates 8 reactors on one site).
    - Instead of building an ultra-high-pressure dome for every single reactor, the individual reactor buildings are kept at a slightly negative pressure and connected via massive relief valves to a central, shared **Vacuum Building**.
    - This massive concrete silo is maintained at a near-absolute vacuum. In the event of a LOCA in any reactor, the overpressure blows the valves open, instantly sucking the radioactive steam out of the reactor building. A dousing tank in the roof of the vacuum building sprays cold water to condense the steam and drop the pressure of the entire complex to safe levels within seconds.

Historical Case Study: The 1983 Pickering Tube Rupture

The theoretical safety advantages of the CANDU's distributed pressure system were put to a real-world test on August 1, 1983, at the Pickering Nuclear Generating Station in Ontario.

The Failure Mechanism: While operating at full power, a 2-meter-long crack suddenly unzipped along pressure tube G16 in Unit 2. The root cause was traced back to the **garter springs** designed to separate the inner pressure tube from the outer calandria tube. A spring had shifted out of place, allowing the heavy, 310°C inner tube to sag and physically contact the cold (70°C) calandria tube. This extreme temperature gradient caused hydrogen and deuterium isotopes in the metal to migrate to the cold contact point, creating brittle **hydride blisters**. The embrittled Zircaloy-2 metal ultimately shattered under the 9 MPa of coolant pressure.

The Safety Case Proven: If a massive PWR pressure vessel had unzipped in this manner, it would have been a catastrophic Large Break LOCA. However, because this was a CANDU:

- It resulted in a manageable **Small Break LOCA**. The heavy water poured into the reactor vault but was entirely contained.
- The outer calandria tube *did not fail*. The massive pool of cold moderator water absorbed the heat, acting as a passive heat sink and preventing any fuel melting.
- The reactor was safely shut down with **zero radioactive release** to the public or injuries to workers.

The Industry Fix & Restart: Following this incident, the industry realized standard Zircaloy-2 was too susceptible to hydride embrittlement. Both Unit 1 and Unit 2 underwent a massive, first-of-its-kind retubing effort to replace every single pressure tube. This unprecedented maintenance kept the reactors offline for years, with Unit 1 finally restarting in September 1987 and Unit 2 in October 1988. Modern CANDUs now strictly use a stronger **Zirconium-2.5% Niobium** alloy for their inner pressure tubes, alongside redesigned, tighter-fitting garter springs.

References and Additional Reading

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